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Chapter 1

Electrostatics

1.1 Overview of Electrostatics

1.1.1 Problem definition of Electrodynamics

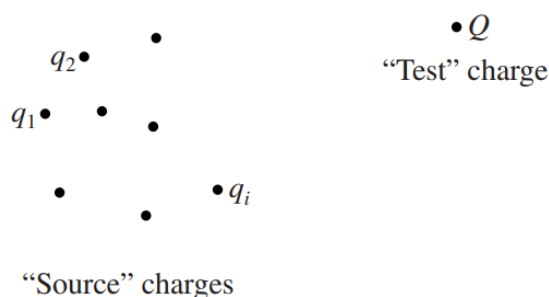


FIGURE 2.1

1. **Problem definition.** We have some electric charges, $q_1, q_2, q_3, \dots$ (**source** charges); what force do they exert on another charge, Q (**test** charge)?
 - **Given.** The positions of the source charges are given (as functions of time).
 - **Goal.** the trajectory of the test particle is to be calculated.
2. **Superposition principle simplifies this problem.** The solution to this problem is facilitated by the **principle of superposition**, which states that the interaction between any two charges is completely unaffected by the presence of others. This means that to determine the force on Q , we can first compute the force $\mathbf{F}_1$, due to q_1 alone (ignoring all the others); then we compute the force $\mathbf{F}_2$, due to q_2 alone; and so on. Finally, we take the mathbftor sum of all these individual forces:

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots$$

Thus, if we can find the force on Q due to a *single* source charge q , we are, in principle, done (the rest is just a question of repeating the same operation over and over, and adding it all up).

3. Difficulty of the problem.

- In general, both the source charges and the test charge are in motion.
- Not only does the force on Q depend on the separation distance $|\mathbf{r} - \mathbf{r}'|$ between the charges, it also depends on both their velocities and on the acceleration of the source charge q .
- Moreover, it is not the position, velocity, and acceleration of q right now that matter: electromagnetic “news” travels at the speed of light, so what concerns Q is the position, velocity, and acceleration that the source charge q had at some earlier time, when the message left.

1.1.2 Time-independent Electrodynamics

1. The study of electrodynamics begins with the special case where source charges are stationary, or electric currents are steady.
2. **The case of electrostatics.** If the source charges, represented by the charge density $\rho(\mathbf{r}, t)$, are stationary,

$$\rho(\mathbf{r}, t) = \rho(\mathbf{r}), \quad \therefore \frac{\partial \rho}{\partial t} = 0,$$

Then the resulting electric field configuration is time-independent, and the Ampere-Maxwell law reduces to

$$\frac{\partial \mathbf{E}}{\partial t} = 0, \quad \therefore \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = \mu_0 \mathbf{J}$$

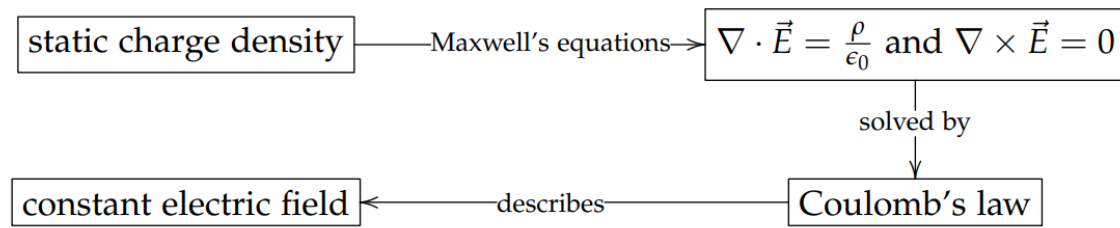
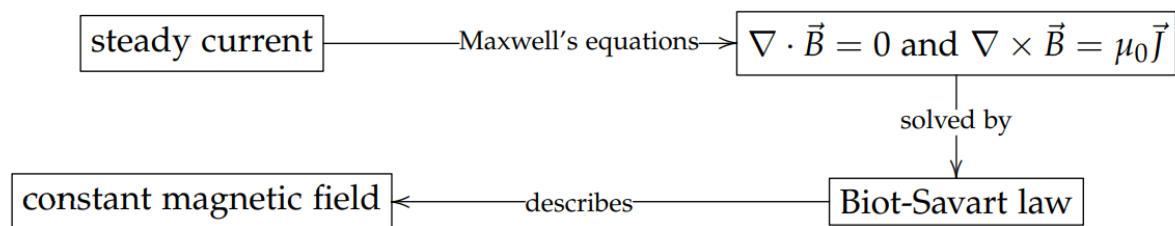
3. **The case of magnetostatics.** If the electric current, represented by the current density $\mathbf{J}(\mathbf{r}, t)$, is steady,

$$\mathbf{J}(\mathbf{r}, t) = \mathbf{J}(\mathbf{r}), \quad \therefore \frac{\partial \mathbf{J}}{\partial t} = 0$$

Then the resulting magnetic field configuration is time-independent, and the Faraday’s law reduces to

$$\frac{\partial \mathbf{B}}{\partial t} = 0, \quad \therefore \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} = 0$$

4. Therefore, in time-independent electrodynamics, Maxwell’s equations become two **separate sets** of equations. In other words, we can treat the electric field and the magnetic field **independently** because only if the electric field is changing does it have an effect on the magnetic field and vice versa. This is summarized in the following page:

Electrostatics**Magnetostatics**

1.1.3 Summary of methods for computing electric fields

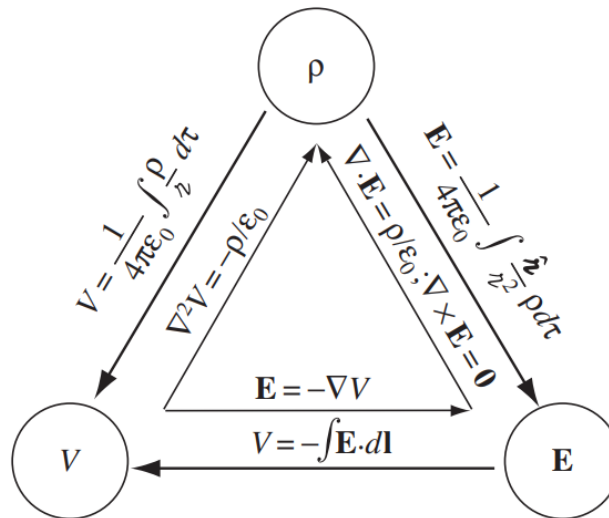


FIGURE 2.35

1. **Motivation.** In the typical electrostatic problem you are given a source charge distribution ρ , and you want to find the electric field $\mathbf{E}$ it produces. Unless the symmetry of the problem allows a solution by Gauss's law, it is generally to your advantage to calculate the potential first, as an intermediate step.
2. These are the three fundamental quantities of electrostatics: ρ , $\mathbf{E}$, and V . All six formulas interrelating them are neatly summarized in Fig. 2.35.
3. Note that we began with just two experimental observations: (1) the *principle of superposition*—a broad general rule applying to *all* electromagnetic forces, and (2) *Coulomb's law*—the fundamental law of electrostatics. From these, all else followed.

1.2 Methods of computing electric fields

1.2.1 Method 1. $\rho \rightarrow \mathbf{E}$: Coulomb's law and superposition principle

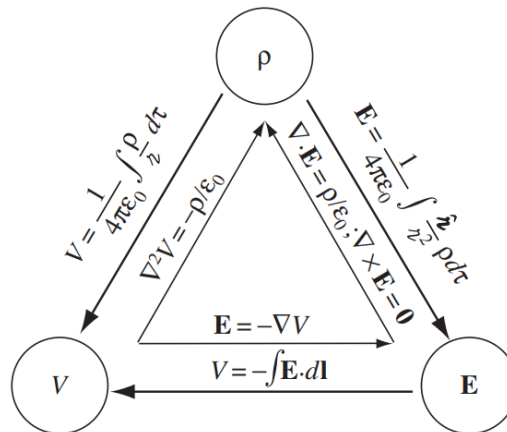


FIGURE 2.35

1. Given a charge distribution $\rho(\mathbf{r})$, the most elementary and brute-force method to compute $\mathbf{E}$ is to exploit Coulomb's law and the principle of superposition.

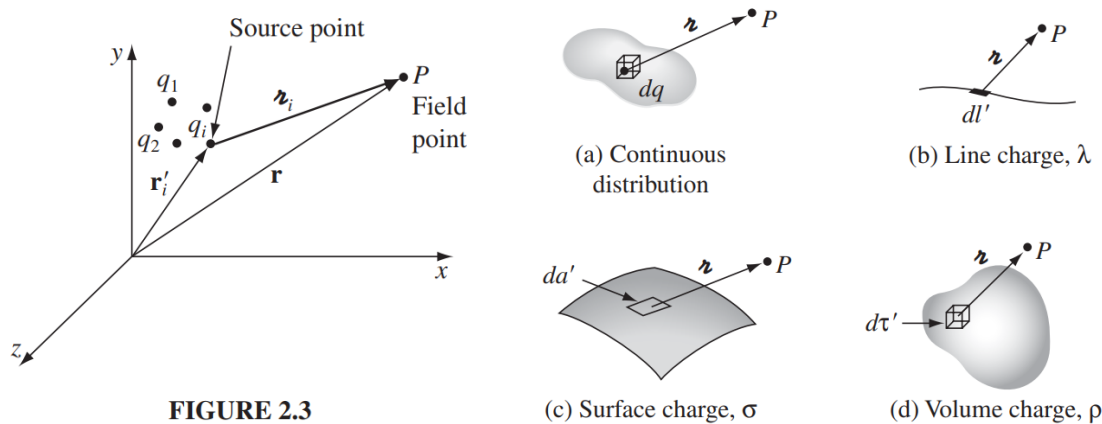


FIGURE 2.3

2. If the source of the field is a collection of **discrete point charges** q_i , the total electric field is simply the sum(superposition) of the electric field produced by each point charge:

$$\mathbf{E}(\mathbf{r}) = \sum_{i=1}^n \mathbf{E}_i = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i (\mathbf{r} - \mathbf{r}'_i)}{|\mathbf{r} - \mathbf{r}'_i|^3}$$

3. If, instead, the charge is distributed continuously over some region, the sum becomes an integral. We integrate(superposition) the infinitesimal electric field $d\mathbf{E}$ produced by the infinitesimal charge element dq as follows:

$$\mathbf{E}(\mathbf{r}) = \int d\mathbf{E} = \int \frac{1}{4\pi\epsilon_0} \frac{dq}{|\mathbf{r} - \mathbf{r}'|^3} (\mathbf{r} - \mathbf{r}')$$

The type of integral depends on the kind of infinitesimal charges classified as follows:

- For a *line charge* with linear charge density λ ,

$$dq = \lambda(\mathbf{r}') dl', \quad \mathbf{E}(\mathbf{r}) = \int \frac{1}{4\pi\epsilon_0} \frac{\lambda(\mathbf{r}') d\mathbf{l}'}{|\mathbf{r} - \mathbf{r}'|^3} (\mathbf{r} - \mathbf{r}')$$

- For a *surface charge* with surface charge density σ ,

$$dq = \sigma(\mathbf{r}') dS', \quad \mathbf{E}(\mathbf{r}) = \int \frac{1}{4\pi\epsilon_0} \frac{\sigma(\mathbf{r}') d\mathbf{S}'}{|\mathbf{r} - \mathbf{r}'|^3} (\mathbf{r} - \mathbf{r}')$$

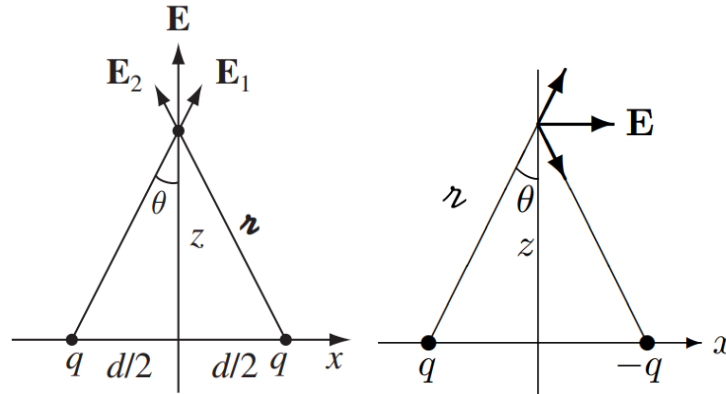
- For a *volume charge* with volume charge density ρ ,

$$dq = \rho(\mathbf{r}') dV', \quad \mathbf{E}(\mathbf{r}) = \int \frac{1}{4\pi\epsilon_0} \frac{\rho(\mathbf{r}') d\mathbf{V}'}{|\mathbf{r} - \mathbf{r}'|^3} (\mathbf{r} - \mathbf{r}').$$

1.2.2 Showcase of electric fields produced by representative configurations

All the results are calculated using Coulomb's law (direct integration).

Case 1. Two point charges of the same(opposite) polarities

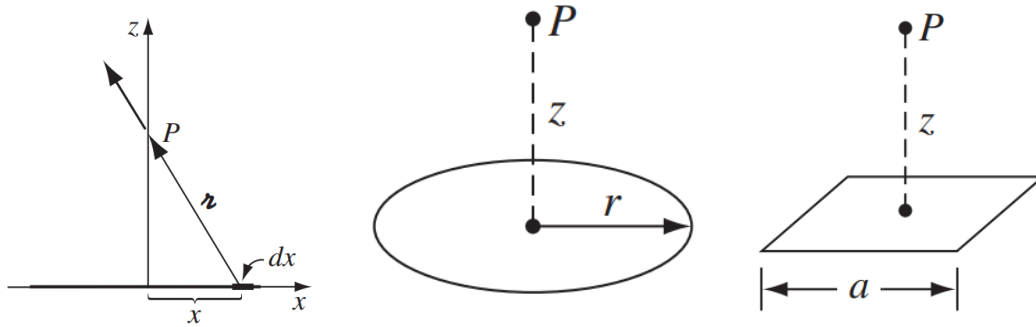


- **Left.** Two point charges of the same polarity.

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{2q}{z^2} \frac{1}{\left[1 + \left(\frac{d}{2z}\right)^2\right]^{3/2}} \hat{z}. \xrightarrow{d \ll z} \frac{1}{4\pi\epsilon_0} \frac{2q}{z^2} \hat{z}$$

- **Right.** Two point charges of opposite polarity.

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{qd}{z^3} \frac{1}{\left[1 + \left(\frac{d}{2z}\right)^2\right]^{3/2}} \hat{x}.$$

Case 2. Line charge distributions with a line charge density λ 

- **Left.** A finite line section of length $2L$

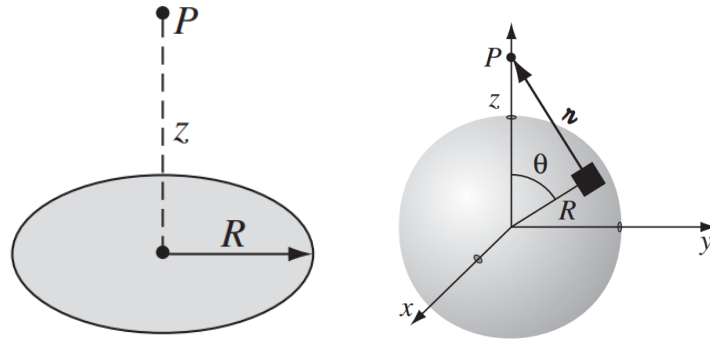
$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{2\lambda L}{z^2} \frac{1}{\sqrt{1 + \left(\frac{L}{z}\right)^2}} \hat{z}$$

- **Middle.** A circle of radius r

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda(2\pi r)}{z^2} \frac{1}{\left[1 + \left(\frac{r}{z}\right)^2\right]^{3/2}} \hat{z}$$

- **Right.** A square of side length a

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \cdot \frac{4\lambda a}{z^2} \frac{1}{\left[1 + \left(\frac{a}{2z}\right)^2\right] \sqrt{1 + \frac{1}{2}\left(\frac{a}{z}\right)^2}} \hat{z}.$$

Case 3. Surface charge distributions with a surface charge density σ 

- **Left. A uniformly charged disk**

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} 2\pi\sigma \left[1 - \frac{1}{\sqrt{1 + \left(\frac{R}{z}\right)^2}} \right] \hat{\mathbf{z}} \xrightarrow{R \ll z} \frac{\sigma}{2\epsilon_0} \hat{\mathbf{z}}$$

- **Right. A uniformly charged spherical shell (or solid sphere)**

$$\mathbf{E}_{out} = \frac{1}{4\pi\epsilon_0} \frac{q}{z^2} \hat{\mathbf{z}} \quad (z > R), \quad \mathbf{E}_{shell} = 0 \quad (z < R), \quad \mathbf{E}_{solid} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} z \hat{\mathbf{z}} \quad (z < R)$$

(Remark. The resulting integration is challenging but doable.)

However, for the spherical charge distribution, z axis can represent any radial direction without loss of generality. In other words, spherical symmetry assures that the electric fields of the spherical charge distribution are all radial:

$$\mathbf{E}_{out} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} \mathbf{r} \quad (r > R), \quad \mathbf{E}_{shell} = 0 \quad (r < R), \quad \mathbf{E}_{solid} = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} \mathbf{r} \quad (z < R)$$